SMART ON FHIR

https://hl7.org/fhir/smart-app-launch/index.html

The Challenge

„EHR are becoming commodity platforms. The winner will be the EHR vendor that provides the *best platform for innovation* – the *most open* and *most extensible* platform.“

-- CEO of major IDN (Integrated Delivery Network)

An App Store for innovative clinical apps that can "plug and play" inside any compliant EHR

SMART®

"Sustainable Medical Applications and Reusable Technology"

“SMART on FHIR” – EHR as Open Platform

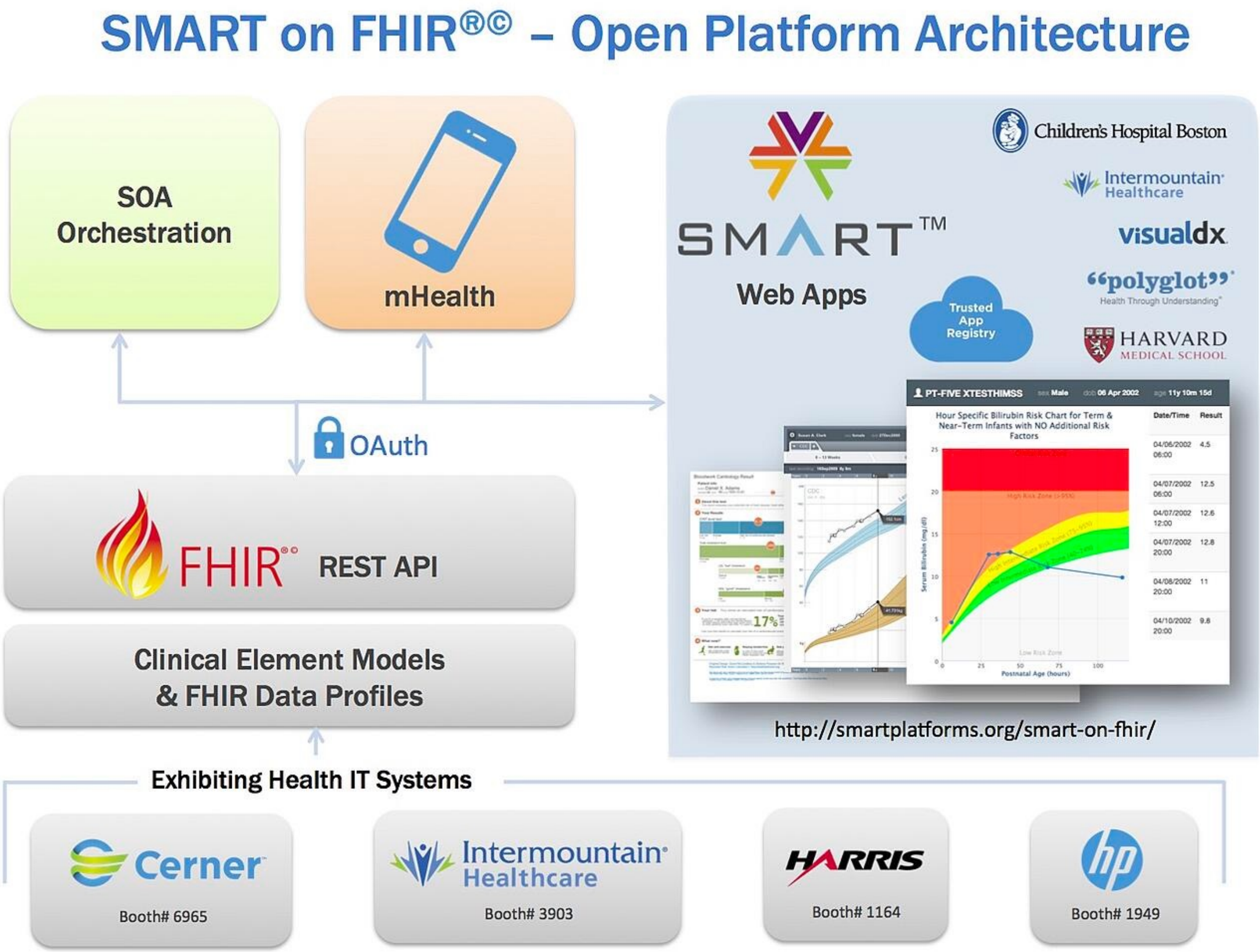
- Modern EHRs are becoming platforms
  - User and patient management
  - Workflow and core services (order, documentation, PAMI (problem, allergy, medication, immunization) data, etc.)
- Use “edge extensions” to complete the functionality
  - App extensions that plug in to the clinical workflow
  - No single vendor can supply every needed function
  - Innovation of single-minded App vendors?
- “App Store” model is now well-understood
  - Many vendors have proprietary APPs for extensions
  - Emergence of <u>robust</u> app market will require standards-based APIs?

SMART

FH | JOANNEUM University of Applied Sciences

- Substitutable Medical Apps Reusable Technologies
- Started as a US research project
- Boston Children's Hospital Computational Health Informatics Program and the Harvard Medical School Department for Biomedical Informatics
- Open Standards
- Aligning with FHIR® standards development
- Funded by US Office of the National Coordinator
- US Argonaut Project:12+ funding vendors; 70+ others

SMART on FHIR



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SMART's Core Focus

Clinicians
Researchers
Consumers
(et al.)

- Consumers want to use info in varied ways
- Health record systems are many and varied

What is SMART about?

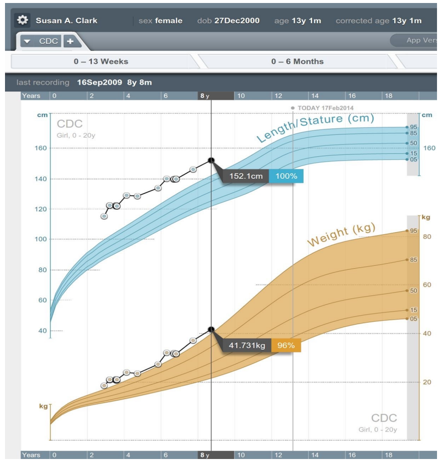
SMART's Core Focus

Clinicians
Researchers
Consumers
(et al.)

Healthcare
IT Systems
>1200 Certified EHRs
& Portals
& Data warehouses

SMART's Core Focus

Apps
Tools, visualisation, What-if?, entry forms, planning etc.

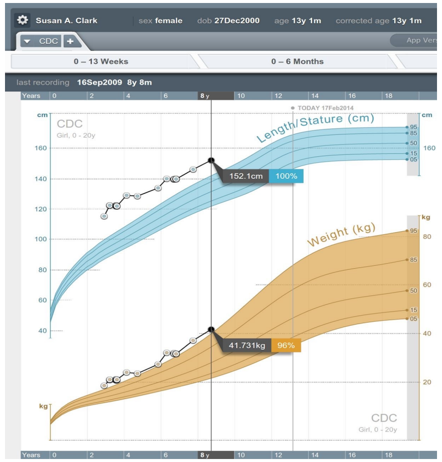


Healthcare IT Systems

- plug-and-play; substitutable
- Need some things:
  - clinical data: format and terminology
  - authentication/authorisation: policy, decisions
  - appropriate access: mobile, web, integrated

SMART's Core Focus

Apps
Tools,
visualisation,
What-if?,
entry forms,
planning etc.



clinical data
authorization
authentication
UX

Healthcare IT Systems

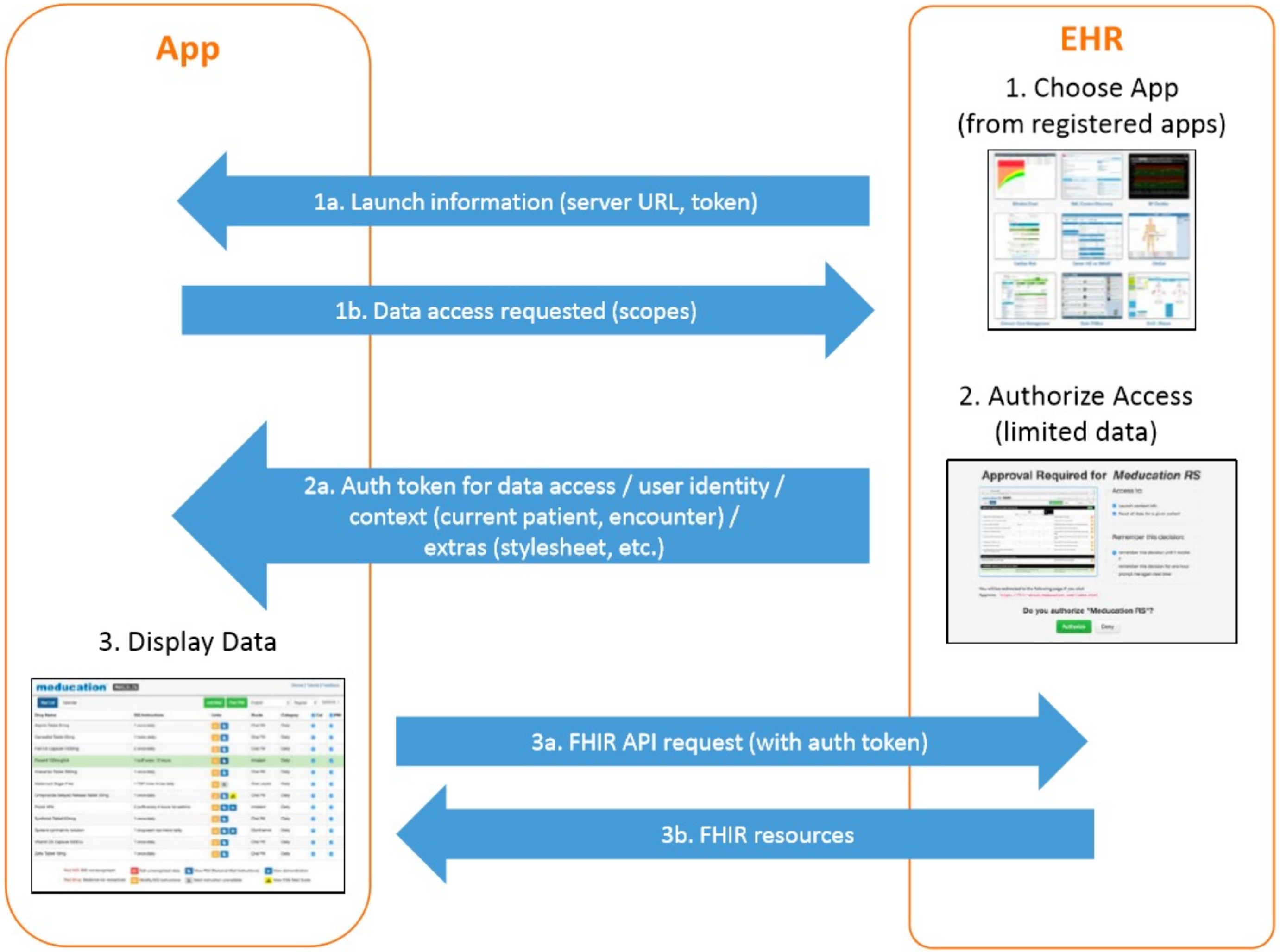
Why SMART?

- Users:
  - App choice (substitutability)
- Developers:
  - Low barriers to entry (open standards, large community)
  - Single app can run in systems by different vendors
  - Single app can run in different contexts (e.g. HER and Patient Portal)

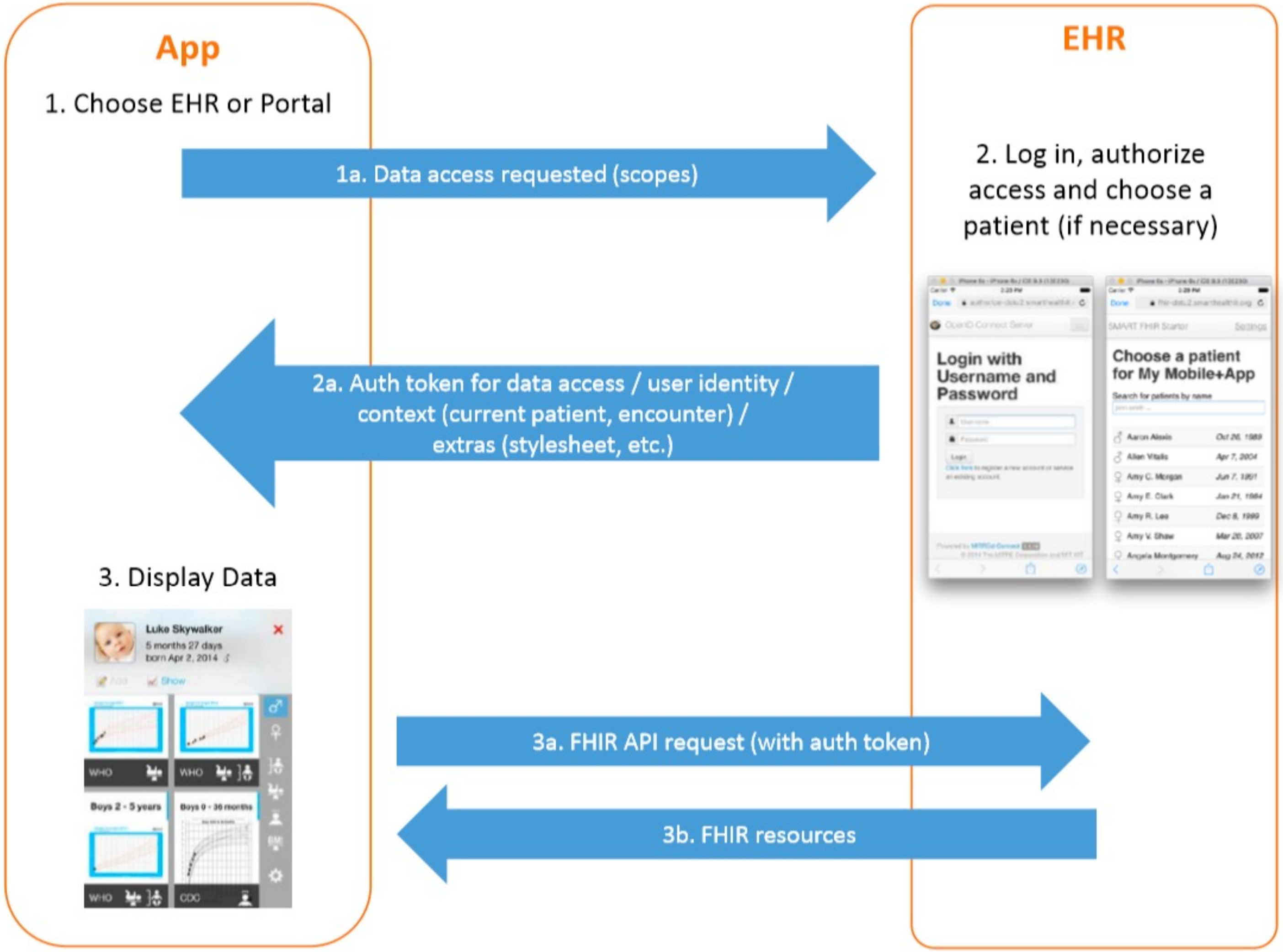
Modern EHRs become a platform!

- User and Patient Management
- Workflow and core services
- Data persistence
- Regulatory compliance
- Apps

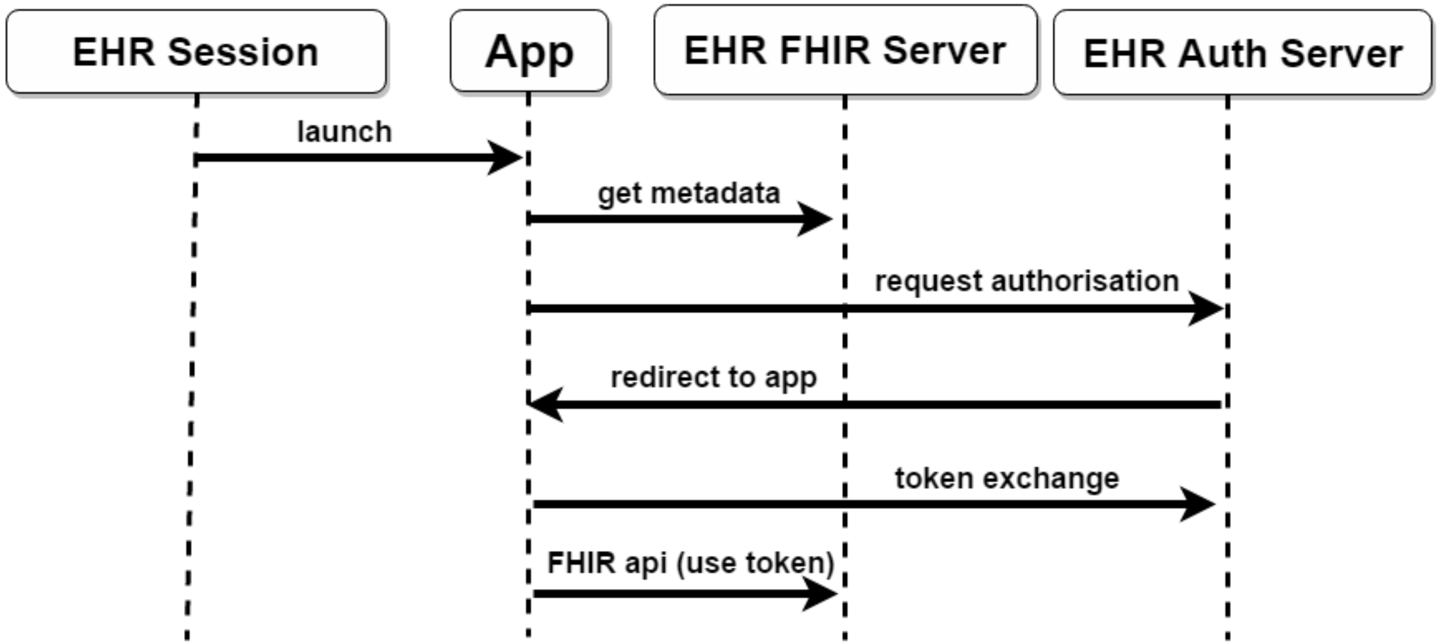
EHR Launch


App
EHR
1. Choose App
(from registered apps)
1a. Launch information (server URL, token)
1b. Data access requested (scopes)
2. Authorize Access
(limited data)
Approval Required for Meducation RS
Do you authorize "Meducation RS"?
2a. Auth token for data access / user identity / context (current patient, encounter) / extras (stylesheet, etc.)
3. Display Data
meducation
3a. FHIR API request (with auth token)
3b. FHIR resources

Standalone Launch



Authentication / Authorisation



- Establish security context (OAuth2)
- Launch an application (supply FHIR server endpoint)
- Locate the EHR Auth Server (Conformance)
- Obtain security token for FHIR server access

SMART Authorization Scopes

- Scopes convey what access an app needs

  patient/Immunization.read

  Access Type

  FHIR Resource

  Permission

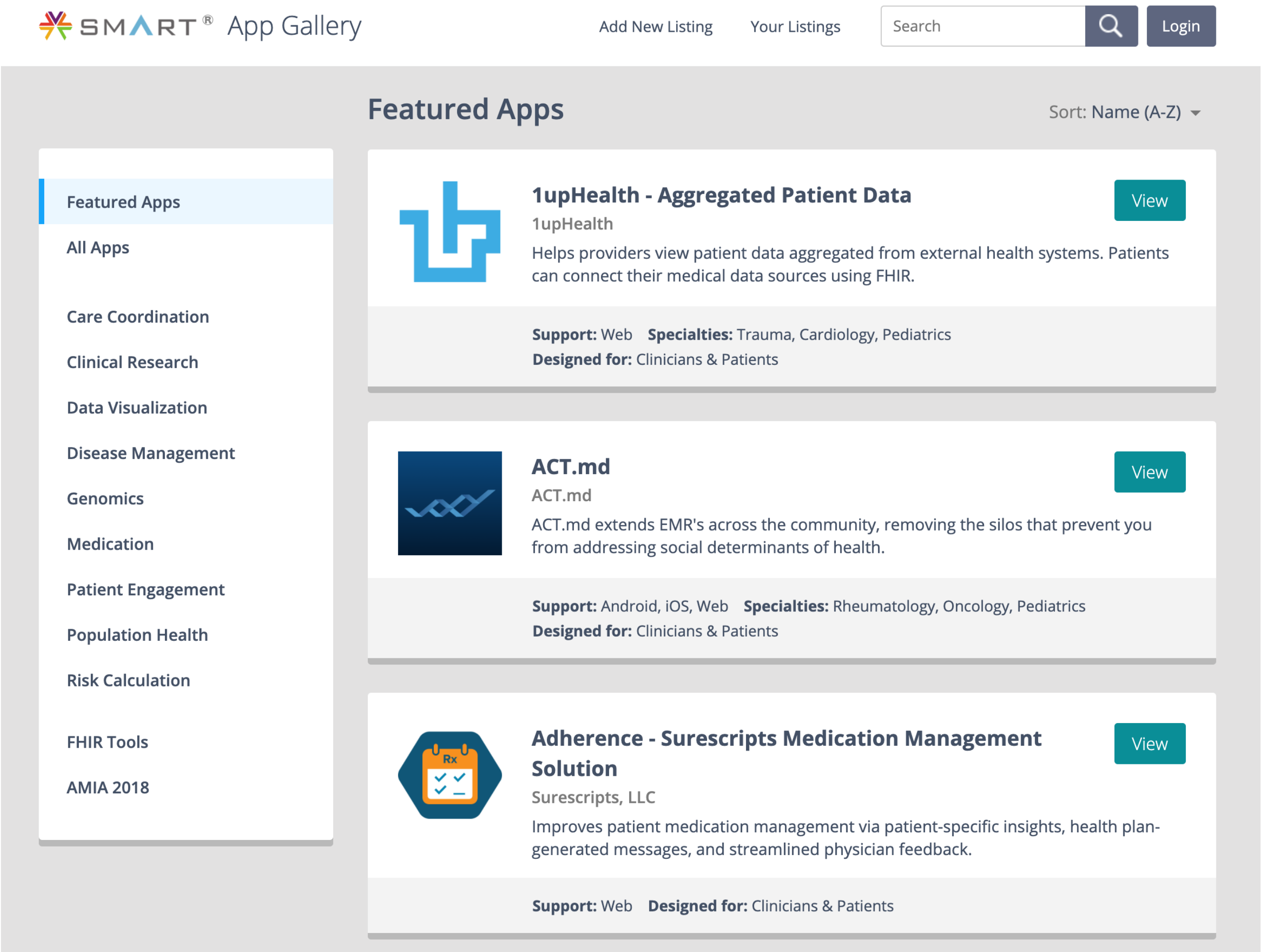
- Examples:
  - Simple app: **patient/Patient.read, patient/Observation.read**
  - Complex app: **patient/*.read**
  - ePrescribing app: **patient/MedicationOrder.write**
  - Population health app: **user/*.read**

http://hl7.org/fhir/smart-app-launch/scopes-and-launch-context/index.html

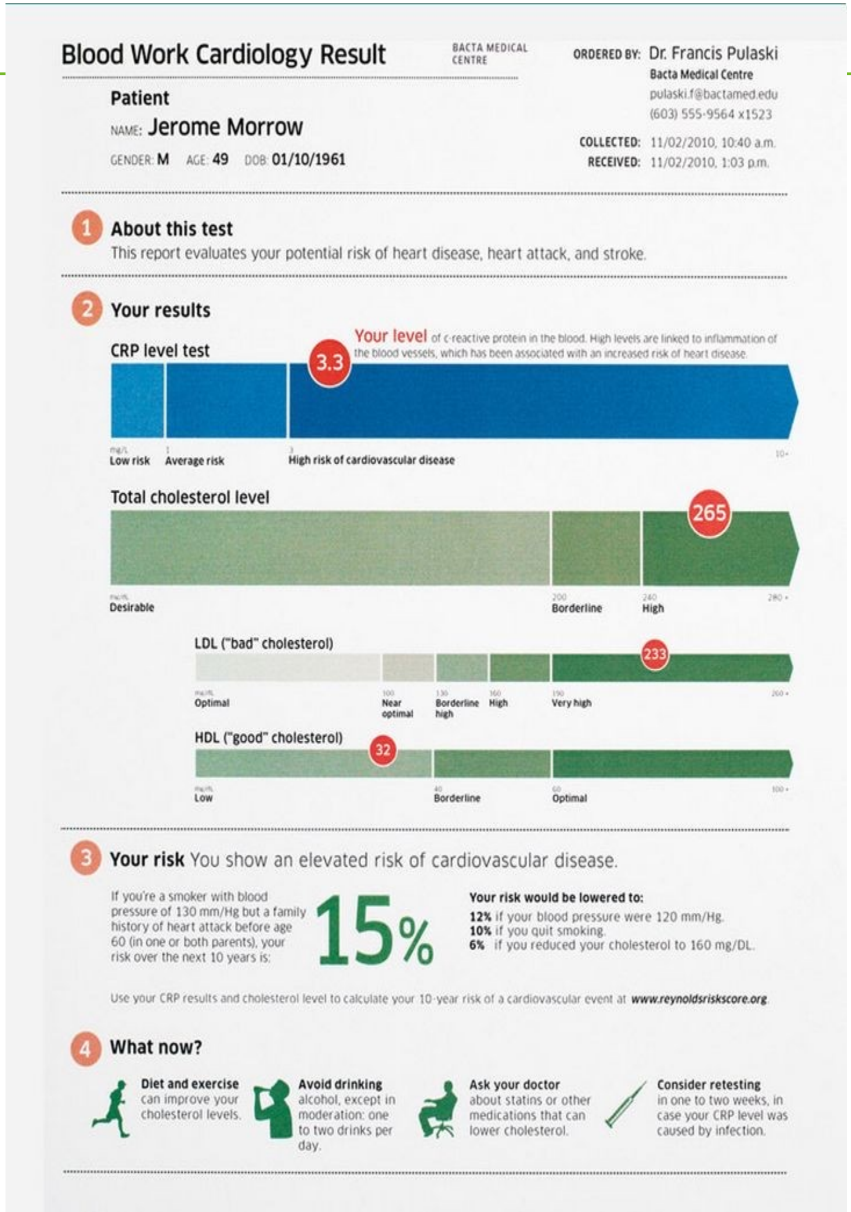
Public App Gallery

- SMART App Gallery offer a single place to find and learn about SMART and FHIR apps
- Vendor and license neutral
  - Not restricted to a single HER platform
  - Hosts commercial and open source apps
- No cost to list or browse apps

- http://apps.smarthealthit.org/



App Example


Blood Work Cardiology Result
BACTA MEDICAL CENTRE

ORDERED BY: Dr. Francis Pulaski
Bacta Medical Centre
pulaski.f@bactamed.edu
(603) 555-9564 x1523

COLLECTED: 11/02/2010, 10:40 a.m.
RECEIVED: 11/02/2010, 1:03 p.m.

Patient
NAME: Jerome Morrow
GENDER: M AGE: 49 DOB: 01/10/1961

1 About this test
This report evaluates your potential risk of heart disease, heart attack, and stroke.

2 Your results

CRP level test
Your level of c-reactive protein in the blood. High levels are linked to inflammation of the blood vessels, which has been associated with an increased risk of heart disease.
3.3
mg/L Low risk
1 Average risk
3 High risk of cardiovascular disease
10+

Total cholesterol level
265
mg/dL Desirable
200 Borderline
240 High
280+

LDL ("bad" cholesterol)
233
mg/dL Optimal
100 Near optimal
130 Borderline high
160 High
190 Very high
200+

HDL ("good" cholesterol)
32
mg/dL Low
40 Borderline
60 Optimal
100+

3 Your risk You show an elevated risk of cardiovascular disease.

If you're a smoker with blood pressure of 130 mm/Hg but a family history of heart attack before age 60 (in one or both parents), your risk over the next 10 years is: 15%

Your risk would be lowered to:
12% if your blood pressure were 120 mm/Hg.
10% if you quit smoking.
6% if you reduced your cholesterol to 160 mg/DL.

Use your CRP results and cholesterol level to calculate your 10-year risk of a cardiovascular event at www.reynoldsriskscore.org

4 What now?

Diet and exercise can improve your cholesterol levels.

Avoid drinking alcohol, except in moderation: one to two drinks per day.

Ask your doctor about statins or other medications that can lower cholesterol.

Consider retesting in one to two weeks, in case your CRP level was caused by infection.

7 best Smart on FHIR Apps:

https://blog.interfaceware.com/top-7-best-smart-fhir-apps/

SMART on FHIR

- https://docs.smarthealthit.org/

SMART®

Standards and Specifications

- Foundational
  - FHIR: Fast Healthcare Interoperability Resources. Web standard for health interop
  - CDS Hooks: Clinical Decision Support Hooks. Web standard for CDS in the EHR workflow
- Data access
  - US Core Data Profiles: FHIR data profiles for health data in the US ("core data for interoperability")
  - FHIR Bulk Data API Implementation Guide: FHIR export API for large-scale data access
- UI and Security Integration
  - SMART App Launch: User-facing apps that connect to EHRs and health portals
  - SMART Backend Services: Server-to-server FHIR connections

Tutorials

- Getting started with Browser-based Apps: Tutorial to create a simple app that launches via the SMART browser library
- Cerner's Browser-based app tutorial: In-depth tutorial to build a simple browser-based app
- Getting started with CDS Hooks: Tutorial to create a simple CDS Hooks Service
- Getting started for EHRs: Tutorial to SMART-enable a clinical data system